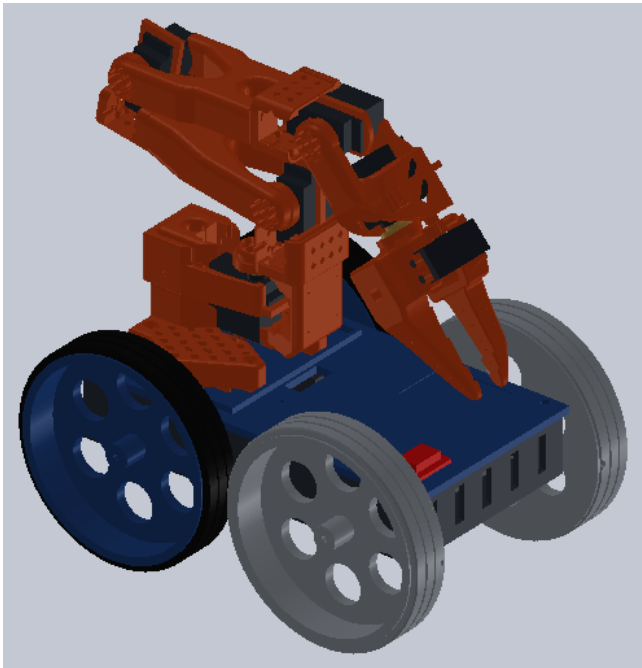


THENAR WALKER

Mobile robot arm for **RoboReach** — Techfest IIT Bombay 2026-27

Team build guide: what we are building, what to buy, how to wire, flash, build and test it.



The robot folded in its start pose (STOW) — SolidWorks render

	Rule	Ours (measured)
Start size	$\leq 300 \times 200 \times 300$ mm	293.7 $\times$ 196.0 $\times$ 292.9 mm
Ground clearance	≥ 50 mm	57 mm
Grasp object centred at	450 mm	449.6 mm
Voltage	≤ 24 V DC	12.6 V max (3S LiPo)
Mass	—	2.39 kg
Drive	—	4 $\times$ 25GA-370 12 V 60 RPM, Ø140 mm wheels, ~0.34 m/s
Full course in simulation	—	pick $\rightarrow$ ramp $\rightarrow$ bridge $\rightarrow$ speed breakers $\rightarrow$ place: PASS

Contents — 2 Rules · 3 Robot at a glance · 4 The arm · 5 Drive & turning · 6–7 Shopping list · 8 Wiring · 9 Firmware & safety · 10 Build guide · 11 Verified in SolidWorks · 12 Simulated run + team checklist

The challenge and the rules that shape the robot

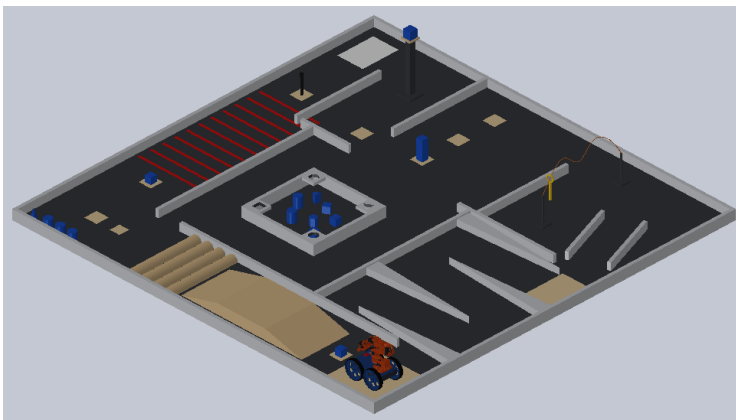
RoboReach: a **manually controlled, wireless** robot with an arm. One operator, 7 minutes, 15 obstacles/tasks on a 3 × 3 m arena. Points + time bonus (300 – time, max 100) – penalties. Max 870 + 100.

Rules that decide the design

Rule	What we did
Fit in 300 × 200 × 300 mm at the start	Arm folds to STOW on a low deck (103 mm) ; robot 293.7 × 196 × 292.9 mm
Grab an object centred at 450 mm, bot on the ground	Arm on the deck reaches ~640 mm; REACH_450 pose puts a cube centre at 449.6 mm
Ground clearance ≥ 50 mm	Ø140 wheels, motors on the axle line → lowest point 57 mm
Wheels always touch the ground; ramp 15°, bumps 40 mm	4WD, big wheels; checked in SolidWorks + physics simulation
Mechanical gripper only	The thenar-arms jaw gripper (no glue, magnets, suction)
Wi-Fi/BT, name TeamName_TF , authentication	ESP32 Wi-Fi AP, every command signed (HMAC-SHA256), replays rejected
Disconnect → motors stop, arm stops, gripper keeps holding	Firmware HOLD state — tested in simulation with a Wi-Fi drop
On-board battery ≤ 24 V, secured, kill switch	3S LiPo in a screwed cage, 15 A fuse, 30 A kill switch on the deck
No autonomous motion	Pure teleoperation: leader arm + joystick → robot

Tasks and points (where the points are)

Task	Pts	Task	Pts	Task	Pts
Pickup (50 mm cube)	10	Stacker (3 cyl + cone)	90	Buzzed (wire loop)	150
Ascent (15° ramp)	30	Longshot (throw cube)	≤100	Diagonals (250 mm track)	20
Bridge (400 mm wide)	10	Robonardo (draw TF)	130	Escape room (4 shapes)	≤100
Descent + speed breakers	60	Downy (cube at 400 mm)	30	Slalom ridge	40
Touchdown (80 mm square)	10	Hanoi'd (tower of Hanoi)	80	The End (stop in zone)	10

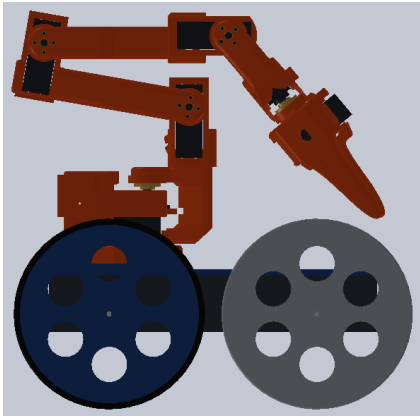


Our SolidWorks model of the 3 × 3 m arena (layout from the rulebook images, ±10 %)

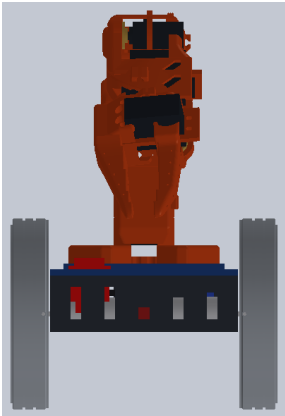
Penalties to avoid: dropping the cube between pickup and touchdown (–15 each), wheels leaving the track (–10), touching the buzz wire (–20 each), letters outside the outline (–30). **Practise Buzzed and Robonardo** — 280 points need a steady arm, not speed.

The robot at a glance

A printed 4WD chassis with the **thenar-arms MG996R follower arm** bolted on top (arm unchanged). The operator uses the **thenar-arms encoder leader** arm plus a thumb joystick.



Side (STOW)

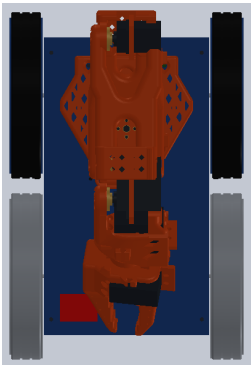


Front (STOW)

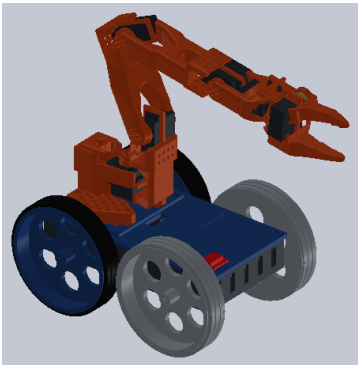
Item	Value	Item	Value
Length × width × height (start)	293.7 × 196 × 292.9 mm	Wheel diameter / width	140 / 28 mm
Wheelbase / track	152 / 168 mm	Ground clearance	57 mm
Chassis tub	240 × 138 mm, printed PETG	Deck top	103 mm
Mass (CAD)	2.39 kg	Centre of gravity height	118 mm (STOW)
Battery	3S LiPo 2200 mAh	Run time per pack	~34 min (run is 7 min)

How it is laid out

- **Tub** (floor 57 mm off the ground): 4 motors screwed to the side walls, battery in a cage on the left, 6 V buck + PCA9685 + 2 motor drivers on the right, 5 V buck at the back. ESP32 hangs under the deck.
- **Deck** (top at 103 mm): the arm base sits in a 4 mm fence and is bolted with 4 × M3 through its own vent slots. Kill switch at the front right.
- **Why the deck is so low:** the arm cannot fold lower than ~190 mm without hitting itself (we searched every pose with collision checking). On a normal 135 mm deck the robot would be 325 mm tall — too tall for the start box.

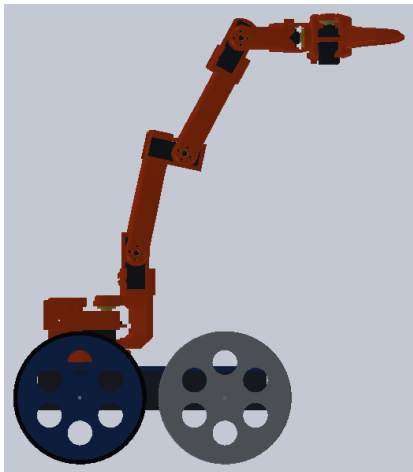


Top view

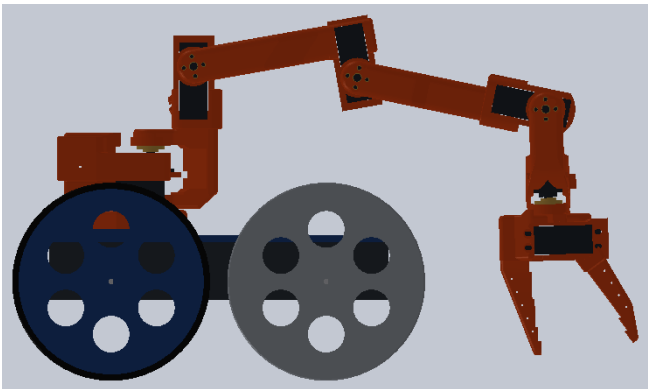


HOME pose (carrying pose)

The arm: poses, reach and gripper



REACH_450 — cube centre at 449.6 mm, jaws closing sideways



PICK_FLOOR — 50 mm cube on the floor

Arm poses (joint angles in degrees: pan, lift, elbow, wrist flex, wrist roll, gripper)

Pose	Joints	Use
STOW	0, -80, 80, 50, 85, 0	start box; fold here before the run
HOME	0, -25, 35, 0, 0, 20	carrying the cube over obstacles (most stable)
REACH_450	-3.4, 12, -73.8, 61.3, -85, 35	objects up to 450 mm (rule), Downy pole top
PICK_FLOOR	0, 80, -68.3, 76.7, 3.4, 35	cubes on the floor, 254 mm ahead of centre

Gripper opening (measured on the real jaw meshes)

Gripper angle	0°	10°	20°	30°	35°	40°	50°	60°
Opening (mm)	20	31	42	53	60	64	72	79

- 50 mm cube: grips around 28°. 60 mm cube / Ø60 cylinder: ~38°. Max ~79 mm.
- The jaws do not fully close (≈20 mm gap at 0°). **Add 4 mm TPU finger pads** so it can hold the Robonardo marker and grip thermocol better (the cube slid ~7 mm along bare fingers in the simulation).
- The URDF "tool point" is on the fixed finger; a held 50 mm object is centred 26.5 mm to the side. All heights here use the true object centre.

Driving rule: carry objects in HOME over the ramp, bridge and speed breakers. Stretched-out poses still pass the stability check, but HOME has the most margin (33 mm on the 15° descent).

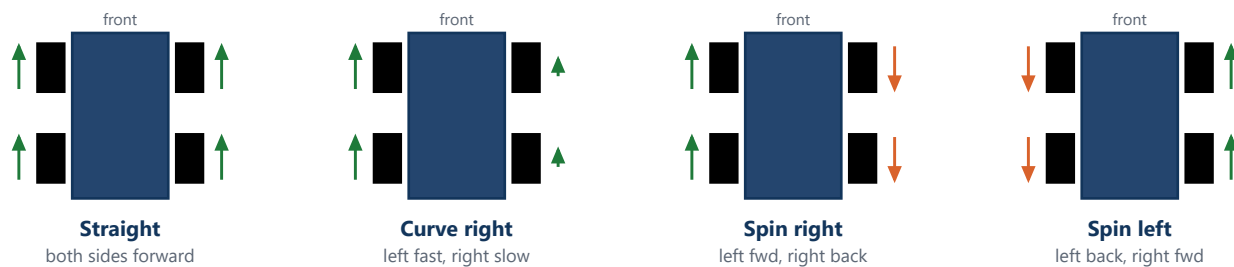
Drive: motors, wheels and turning

Motor to buy		Drive numbers	
Name in shops	25GA-370 12 V 60 RPM (same as JGA25-370 103:1)	Ramp force (15°)	9.16 N
How many	4 (+1 spare)	Needed per motor	1.63 kg·cm (4 driving)
Where	Robu.in ₹389 (in stock when checked); The Engineer Store ₹462	Worst case	3.27 kg·cm (2 driving, on a bump)
Check	12 V · 60 RPM · 4 mm D-shaft · metal gears · no encoder	Motor rated / stall	2.7 / 8.2 kg·cm
Do NOT buy	100 RPM version — too weak for the ramp	Speed	0.34 m/s loaded, 0.44 free

Motor check when it arrives (10 min): put a printed wheel on one motor, tie a string to the tyre, hang a full **1 L water bottle**, run at 12 V. It must lift it. Shops list different strengths for the same name — the simulation needs ≥ 6 kg·cm (5 gets stuck on the speed breakers). If it lifts it, buy the other three.

How it steers — no steering servo, no steering shaft

Skid steering, like a tank or a JCB. One Cytron MDD10A drives both left wheels (channel 1) and both right wheels (channel 2). The ESP32 mixes the joystick: **left = throttle – turn**, **right = throttle + turn**. Different speeds on the two sides turn the robot; opposite directions spin it on the spot. Each wheel is fixed straight on its motor shaft.



Turning — why the front tyres are hard plastic

The robot is almost square (wheelbase 152 $\approx$ track 168), so every turn drags the tyres sideways. With rubbery TPU on all four wheels it could hardly turn. We tested options in the physics simulation:

Variant	Spin 35%	Spin 65%	Spin 100%	Arc turn	Top speed	Ramp
All TPU tyres (first design)	0.2°/s	31.6°/s	83.6°/s	2.8°/s	0.53 m/s	yes
Stronger, slower motor, all TPU	7.5°/s	32.1°/s	64.4°/s	5.3°/s	0.32 m/s	yes
Strongest motor, all TPU	8.2°/s	26.5°/s	52.8°/s	6.3°/s	0.24 m/s	yes
PETG front + TPU rear (chosen)	20.0°/s	77.0°/s	136.2°/s	14.3°/s	0.53 m/s	yes

Study done with a 77 rpm motor. The 60 RPM motor you buy was then re-checked in the full simulation: it spins 145° to within 0.2° and passes the whole course, a bit slower (~ 0.34 m/s).

- **Front wheels: one-piece hard PETG. Rear wheels: PETG hub + TPU 95A tyre.** Hard rear tyres were also tried: they get stuck on the speed breakers.
- Firmware gives turning its own power (50/75/100 %) with a 35 % minimum so a small stick push still turns.
- **Drive like a car** (curves while moving) where you can; spin on the spot for small corrections. In the 250 mm Diagonals corridor the robot (196 mm wide) cannot spin around — plan arcs.

Shopping list (1/2) — rover module

Everything for the driving base. Specs matter more than brand — check against the "Spec" column.

Drive

Qty	Part	Spec / notes	Where to buy
4 (+1)	25GA-370 12 V 60 RPM gear motor	4 mm D-shaft, metal gears, no encoder. Do the 1 L bottle test.	Robu.in (₹389), The Engineer Store
1	Cytron MDD10A dual motor driver	10 A per channel, PWM + DIR, 84.5 × 62 mm. Already bought; sits on the deck top.	bought locally
4	M3 × 5 grub screw (cup point)	Wheel hub onto the motor shaft flat	Local hardware / Robu
8	M3 × 6 socket screw	Motor face to chassis wall (2 per motor)	Local hardware
2 m	22 AWG silicone wire (red/black)	Motor leads	Robu / Amazon.in

Power and safety

Qty	Part	Spec / notes	Where to buy
1 (+1)	3S LiPo 2200 mAh 30C , XT60	Max 106 × 34 × 26 mm to fit the cage	Robu / Amazon.in / hobby shop
1	LiPo balance charger + LiPo-safe bag	Charge only in the bag, never unattended	Robu / Amazon.in
1	KCD4 30 A DPST rocker (red)	The kill switch. Panel cut-out 22.2 × 30.2 mm	Robu / Amazon.in
1	Inline blade fuse holder + 15 A fuse	Right after the battery	Car accessories shop
1	XT60 male/female pair	Battery connector	Robu
1	6 V DC-DC buck, ≥ 10 A	XL4016 / "300 W 20 A" type, adjustable. Feeds the 6 servos	Robu / Amazon.in
1	MP1584 mini buck	Set to 5.0 V for the ESP32	Robu
1 m	16 AWG silicone wire	Battery → fuse → switch → drivers	Robu

Brains

Qty	Part	Spec / notes	Where to buy
1	ESP32-DevKitC (WROOM-32E, 38 pin)	The robot controller (Wi-Fi AP + motors + servos)	Robu / Amazon.in
1	PCA9685 16-ch PWM board	Drives the 6 arm servos over I ² C	Robu
1 + 1	Resistors 100 kΩ + 27 kΩ	Battery voltage sense	Any
1 lot	Jumper wires, servo extensions, heat-shrink, zip ties		Any

Rover prints — 5 ready P1S plates in cad/print, no supports (~29 h total)

Qty	Part	Spec / notes	Where to buy
1	Plate 1: tub + battery cage + 2 bars	PETG, 4 walls, 30 % gyroid (~7 h, ~260 g)	P1S
1	Plate 2: deck plate	PETG, 250 mm long — fits the P1S bed (~3 h, ~130 g)	P1S
2	Plates 3 + 4: front wheel + rear hub	PETG, ~5.4 h each; front wheels are one piece (hard tread). Print plate 3 first (motor test)	P1S
1	Plate 5: 2 rear tyres	TPU 95A , 3 walls, 20 % (~7.7 h, ~90 g)	P1S

Shopping list (2/2) — arm, controller, hardware

Arm — the thenar-arms MG996R follower (unchanged; full list in thenar-arms R3-PRINT.md)

Qty	Part	Spec / notes	Where to buy
6	MG996R servo (180°)	Pan, lift, elbow, wrist flex, wrist roll, gripper	Robu / Amazon.in
6	Metal servo horn 25T, Ø20 mm disc	4 × M3 on 14 mm circle — must match your servo spline	Robu
24 / 24 / 48	M3 × 12 screws / nuts / washers	Servo tabs	Local hardware
24	M3 × 8 screws	Horns to links	Local hardware
7	Arm prints (R3)	Base, shoulder, upper arm, forearm, wrist, gripper body, jaw — PETG	Print
2	TPU finger pads, 4 mm	Recommended (marker + grip)	Print
4	M3 × 20 + washer + nyloc	Arm base to deck, through its vent slots	Local hardware

Controller — added to the thenar-arms encoder leader

Qty	Part	Spec / notes	Where to buy
1	Leader arm (thenar-arms L1)	6 × AS5600, TCA9548A, ESP32, bearings — its own BOM	Already designed
1	2-axis thumb joystick module	3.3 V; VRx → GPIO34, VRy → GPIO35	Robu
1	Toggle switch (ENABLE)	GPIO32 to GND	Robu
1	Push button (SPEED)	GPIO33 to GND	Robu
1	USB power bank + cable	Powers the leader ESP32	Any

Chassis hardware

Qty	Part	Spec / notes	Where to buy
4	M3 heat-set insert (Ø4.0 bore)	Tub corner columns	Robu / Amazon.in
4	M3 × 10 socket screw	Deck to tub	Local hardware
2	M3 × 8 + nyloc	Battery cage to floor	Local hardware
4	M3 × 8 self-tapping	Battery bars	Local hardware
20	M2.5 × 6 self-tapping	Boards to the floor standoffs	Local hardware
2	M2.5 × 10 brass standoff (F-F)	ESP32 under the deck	Robu

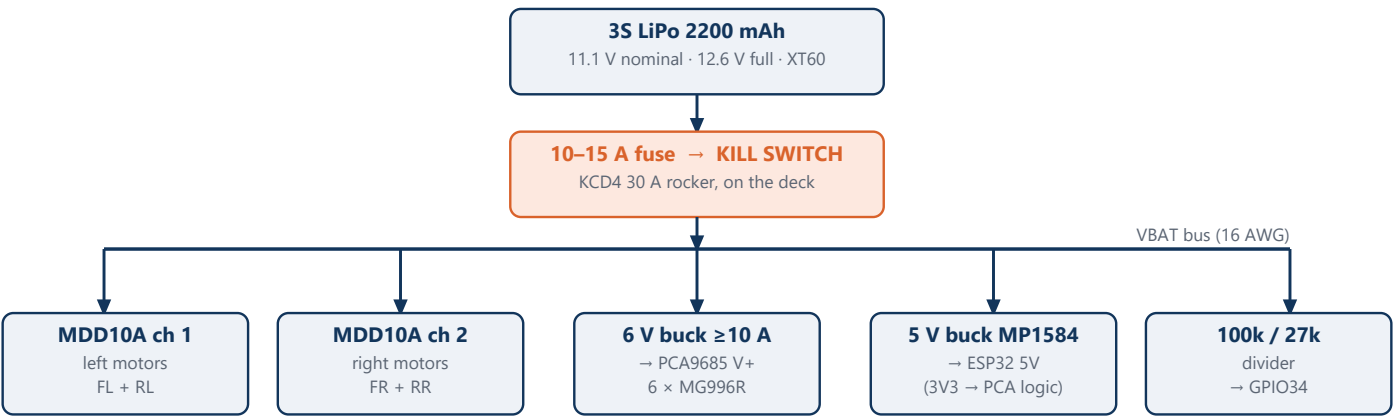
Filament and tools

Qty	Part	Spec / notes	Where to buy
~0.81 kg	PETG	Rover plates 1–4 (+ the arm parts if not printed yet)	
~87 g	TPU 95A	Rear tyres (+ finger pads)	
—	Soldering iron, heat-set insert tip, multimeter, hex keys, crimper	Multimeter is a must for setting the bucks	

Not needed: camera, Raspberry Pi, encoders, IMU. The ESP32 is the right brain here: boots in under a second, survives the kill switch (no SD card), and the safety firmware is already written and tested. Add an ESP32-CAM later only if you want a camera view.

Wiring

Power



All grounds common. Every voltage on the robot ≤ 12.6 V (rule: ≤ 24 V DC).
Kill switch is before everything → it shuts the whole robot down (rule).

Robot ESP32 pins

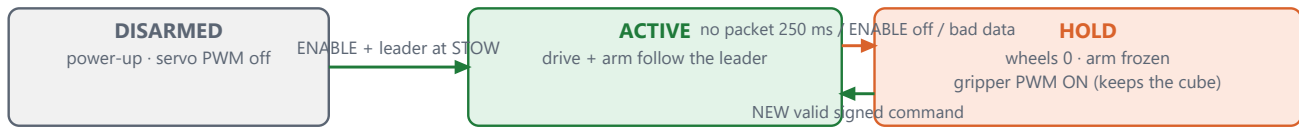
ESP32 pin	Goes to	ESP32 pin	Goes to
GPIO21 / 22	PCA9685 SDA / SCL	GPIO16 / 17	MDD10A PWM2 / DIR2 (right motors)
GPIO13	PCA9685 OE (servo output enable)	GND	MDD10A header GND (pin 1)
GPIO25 / 26	MDD10A PWM1 / DIR1 (left motors)	GPIO34	battery divider (100k / 27k)
GPIO2	on-board LED (solid = ACTIVE)		
5V / GND	5 V buck output	3V3	PCA9685 VCC (logic only)

PCA9685 ch	Servo	Controller (leader ESP32)	
0	J1 shoulder pan	GPIO34 / 35	joystick X (turn) / Y (throttle)
1	J2 shoulder lift	GPIO32	ENABLE toggle → GND
2	J3 elbow	GPIO33	SPEED button → GND
3	J4 wrist flex	3V3 / GND	joystick power (not 5 V)
4	J5 wrist roll	21 / 22	leader encoders (unchanged)
5	J6 gripper		

Before connecting: set the 6 V buck to 6.0 V and the 5 V buck to 5.0 V with a multimeter, no load. Never power servos from the ESP32 or USB. MDD10A: M1A/M1B = both left motors in parallel, M2A/M2B = both right; 10 kΩ from PWM1 and PWM2 to GND. Whole side backwards: swap that side's A/B wires. One wheel fighting its partner: swap that motor's two wires.

Firmware, radio link and safety

Two ESP32 programs share one library (ThenarLink). The safety logic is unit-tested on a PC (39/39 checks) and was run inside the physics simulation.



The kill switch cuts power in every state.

What judges will check — and how we meet it

Rule	How
Name TeamName_TF	Robot makes a WPA2 Wi-Fi network called <TeamName>_TF (spaces removed, max 20 chars), 1 device max
Only our controller	Every packet signed with a 32-byte secret key (HMAC-SHA256); new session id per pairing; old/copied packets rejected
Disconnect	250 ms without a valid packet → HOLD: wheels stop at once, arm freezes, gripper keeps holding
Stays disabled	Moves again only on a NEW valid command; arm catches up slowly (45°/s) for 1.5 s
Kill switch	Hardware, in series with the battery — cuts everything

Driving controls

ENABLE switch on = robot may move. SPEED button cycles **precision 35 % / normal 65 % / fast 100 %** for driving (turning 50 / 75 / 100 %). Power-up: the arm only engages when the leader arm is folded to STOW (within 6°), so it can never jump.

Set up and flash

Step	Command / action
1. Make your secret key	python firmware/tools/new_key.py "Your Team Name"
2. Test the logic on a PC	powershell -File firmware/tests/run_tests.ps1 (39 checks)
3. Compile both sketches	powershell -File firmware/tools/compile_firmware.ps1
4. Flash	thenar_walker_bot → robot ESP32; thenar_walker_controller → leader ESP32
5. Leader calibration	Hold leader at HOME, send ZERO in the serial monitor; SIGN <axis> -1 for reversed joints; JOYZERO
6. Servo calibration	Measure each servo zero/direction with horns off → calibration.h → FOLLOWER_CALIBRATED = true

Until servo calibration is done, the robot drives but the arm stays limp — on purpose. Never share or commit team_config.h.

Build guide

0 · Before buying everything

- ☐ Buy one motor first and do the 1 L bottle test (page 5).
- ☐ Measure motor and boards against the CAD; change cad/tools/design.py if different.

1 · Print

- ☐ The 5 plates in cad/print (PETG plates 1–4, TPU plate 5) — plate 3 first. No supports. Plus the 7 arm parts.

2 · Tub

- ☐ Melt 4 M3 inserts into the corner columns.
- ☐ Screw the battery cage to the floor.
- ☐ Drop motors into the U-slots, gearbox face on the wall, $2 \times M3 \times 6$ each (solder leads first).
- ☐ Screw the 6 V buck, PCA9685 and the 5 V buck to the floor standoffs (2 pads stay spare).
- ☐ Set bucks to 6.0 V / 5.0 V. Wire per page 8, kill switch OFF.

3 · Wheels

- ☐ TPU tyres onto the 2 rear hubs; the 2 front wheels are one piece. Push onto the D-shaft ~1 mm off the wall, grub screw on the flat.

4 · Deck and arm

- ☐ Screw 2 brass standoffs under the deck and zip-tie the ESP32 to them, snap in the kill switch, route servo leads through the slot. MDD10A onto the 4 deck-top bosses, terminals forward, wires down the front slot.
- ☐ Battery into the cage, screw on the bars. Deck onto the tub with $4 \times M3 \times 10$.
- ☐ Arm base into the fence, $4 \times M3 \times 20$ through the base vent slots, nylocs under the deck. Servos J1..J6 → PCA ch 0..5.

5 · Firmware

- ☐ Key, compile, flash, calibrate — page 9.

6 · First power-up (wheels off the ground)

- ☐ Robot and leader in STOW, ENABLE off, kill switch on.
- ☐ ENABLE on → LED solid. Check every wheel direction.
- ☐ **Failsafe test:** while driving and holding a cube, switch the controller off → wheels stop, arm freezes, cube stays in the gripper. Repeat this at inspection.

7 · Before every run

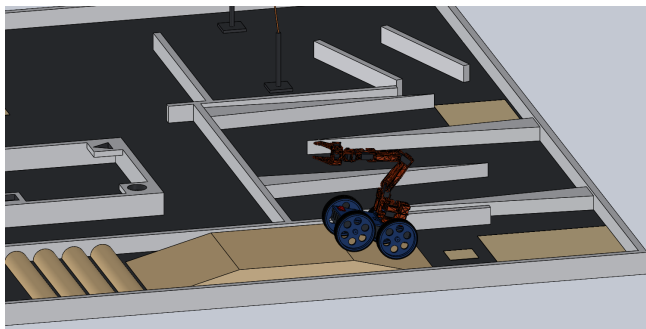
- ☐ Battery ≥ 11.4 V, team key is not the example, robot folded in STOW and inside the $300 \times 200 \times 300$ box.

Verified in SolidWorks

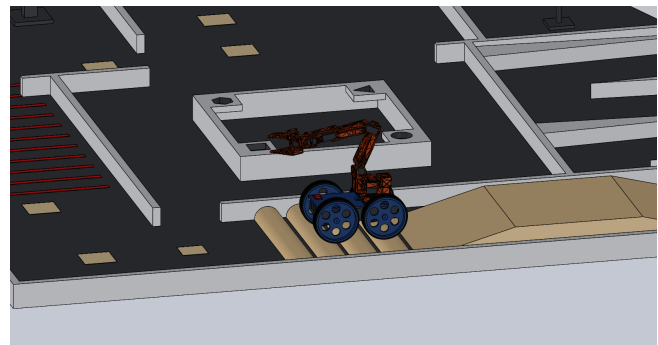
Native SolidWorks 2026 parts and assemblies, generated by scripts. Checks use exact geometry (body extreme points, interference detection). The arm matches the URDF kinematics to 0.04 mm.

Result	Check	Measured
PASS	Start box 300 x 200 x 300 (STOW)	293.7 x 196.0 x 292.9 mm
PASS	Ground clearance $\geq$ 50 mm (STOW)	57.0 mm at chassis tub
PASS	Ground clearance $\geq$ 50 mm (HOME)	57.0 mm at chassis tub
PASS	Ground clearance $\geq$ 50 mm (REACH_450)	57.0 mm at chassis tub
PASS	Grasp object centred at 450 mm (REACH_450)	50 mm object centre z = 449.6 mm, 140 mm ahead of the chassis
PASS	Floor pick: 50 mm cube on the ground ahead of the wheels	grasp centre [254.4, -0.2, 25.1] (cube centre z = 25)
PASS	No interference (STOW)	clear
PASS	No interference (HOME)	clear
PASS	No interference (REACH_450)	clear
PASS	No interference (PICK_FLOOR)	clear
PASS	Static stability (STOW, flat + 15 deg ramp)	worst margin 36.3 mm
PASS	Static stability (HOME, flat + 15 deg ramp)	worst margin 32.9 mm
PASS	Static stability (REACH_450, flat + 15 deg ramp)	worst margin 24.5 mm
PASS	Static stability (PICK_FLOOR, flat + 15 deg ramp)	worst margin 20.5 mm

Robot placed on the arena's real ramp, bridge and speed breakers at 9 positions: **only the tyres touch** — the chassis, motors and arm never hit an obstacle.



On the 15° ramp



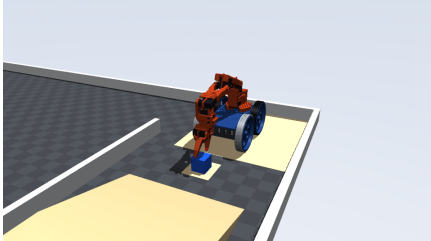
Speed breaker under the belly

Simulated end-to-end run

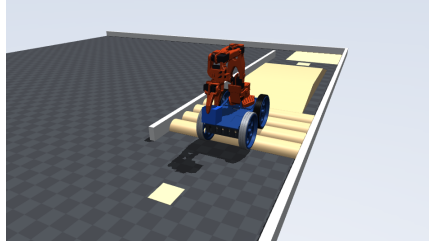
MuJoCo physics (real masses, motor torque curve, servo limits, tyre friction) with **our actual firmware logic in the loop**. A scripted "operator" drives the leader arm and joystick like a person would.

Mission	What	Result	Time
A	Pick 50 mm cube → 15° ramp → bridge → down → 4 speed breakers → place in 80 mm square	PASS 7/7	28 s
B	Same + 1.5 s Wi-Fi drop on the bridge + 75 fake/replayed packets	PASS 12/12	29 s
C	Downy: 60 mm cube off the 400 mm pole, turn, place in its square	PASS 5/5	55 s

Wi-Fi drop: speed 0 mm/s at end, rolled 11 mm after HOLD; max joint drift 0.21 deg; attacks: 75 attack packets, 0 accepted.



Picking the cube



Over the speed breakers, cube held



Cube taken off the 400 mm pole

What the simulation does NOT prove

Tyre and gripper friction are estimates; real thermocol dents (squeeze gently); the scripted operator never shakes. Videos: [sim/out/mission_A.mp4](#), [mission_B.mp4](#), [mission_C.mp4](#).

Team checklist

- ☐ Order 1 motor → bottle test → order the rest + all electronics (pages 6–7)
- ☐ Print the 5 rover plates (cad/print, plate 3 first) and the arm
- ☐ Build and wire (pages 8, 10); set both bucks before connecting
- ☐ Flash, calibrate the leader and the servos (page 9)
- ☐ Test on the real floor: turning, the ramp, a 40 mm bump, the failsafe
- ☐ Practise: Buzzed (150) and Robonardo (130) are a third of all points
- ☐ Register the team; the SSID must be YourTeamName_TF